

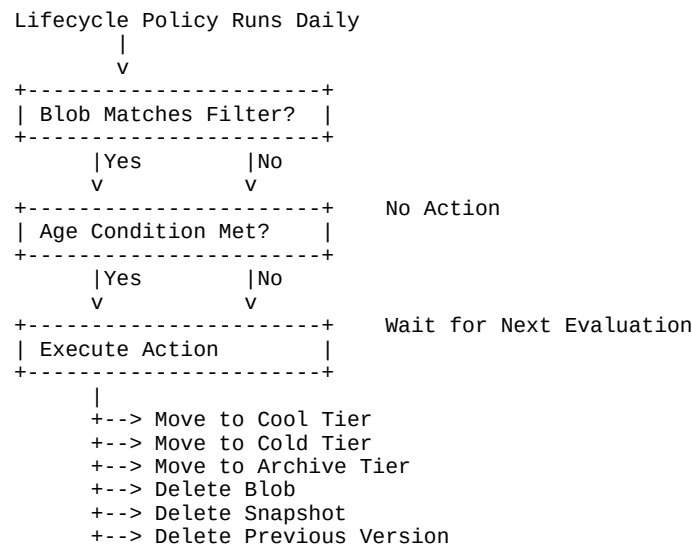
Azure Blob Storage Lifecycle Management

Modern Architecture & Implementation Guide

Overview

Azure Blob Storage Lifecycle Management automatically transitions or deletes blobs based on policies you define. Lifecycle rules help reduce storage costs while maintaining retention and compliance requirements.

How Lifecycle Management Works



Rule Components

- 1 Rule Definition: Name, status, and blob types.
- 2 Filters: Prefix paths and blob index tags.
- 3 Actions: Tiering and deletion operations based on age.

Example Lifecycle Timeline

```
Day 0    -> Blob Created
Day 30   -> Move to Cool Tier
Day 90   -> Move to Archive Tier
Day 365  -> Delete Blob
```

Storage Tiers

Hot Tier: Frequently accessed data.
Cool Tier: Short-term backups and logs.
Cold Tier: Infrequently accessed datasets.
Archive Tier: Long-term retention and compliance storage.

Terraform Example

```
resource "azurerm_storage_management_policy" "lifecycle_policy" {
  storage_account_id = azurerm_storage_account.example.id

  rule {
    name      = "ArchiveAndCleanupWorkflow"
    enabled   = true

    filters {
      prefix_match = ["telemetry-data/logs/", "temp-exports/"]
      blob_types   = ["blockBlob"]
    }

    actions {
      base_blob {
        tier_to_cool_after_days_since_modification_greater_than = 30
        tier_to_archive_after_days_since_modification_greater_than = 90
        delete_after_days_since_modification_greater_than         = 365
      }

      snapshot {
        delete_after_days_since_creation_greater_than = 30
      }
    }
  }
}
```

Best Practices

- 1 Use filters to limit policy scope.
- 2 Validate lifecycle policies in non-production environments.
- 3 Delete snapshots and previous versions when long-term retention is not required.
- 4 Monitor storage costs and access patterns regularly.
- 5 Use Infrastructure as Code for repeatable deployments.